A+ Core 1

Exam 220-1201

A+ 220 1201

Module 8

Summarizing Virtualization and Cloud Concepts

Lesson 8.1

Client-Side Virtualization

Learning Outcomes

What are the differences between Type 1 and Type 2 hypervisors, and which would be more suitable for client-side virtualization on employee workstations?

How can client-side virtualization be used to support legacy software and provide sandbox environments for testing?

What are the key resource requirements for setting up a client-side virtualization workstation?

Learning Outcomes

How does the amount of system memory affect the performance of virtual machines on a client-side virtualization setup?

How does container virtualization differ from traditional virtual machine (VM) virtualization, and what are the benefits of using containers?

Client-Side Virtualization

- Hypervisors
- Uses for Virtualization
- Virtualization Resources Requirements
- Virtualization Security Requirements

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Hypervisors

Hypervisor functions

Allows computers to run multiple OSs on one system for virtualization

Manages virtual machines (VM) by emulating hardware

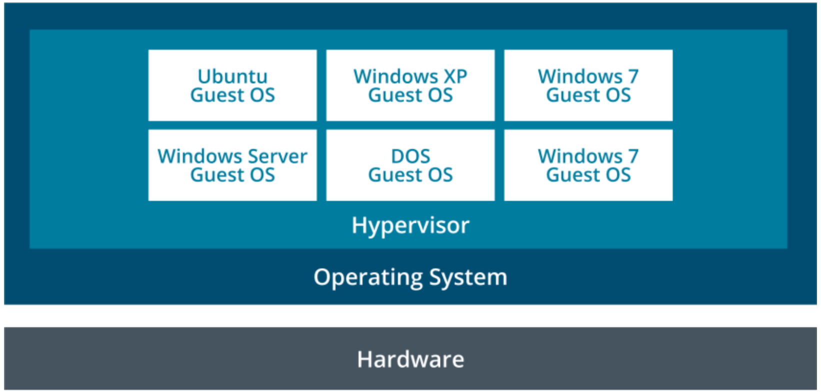
Two main types

Type 2 (Host-based): installed on top of existing host OS

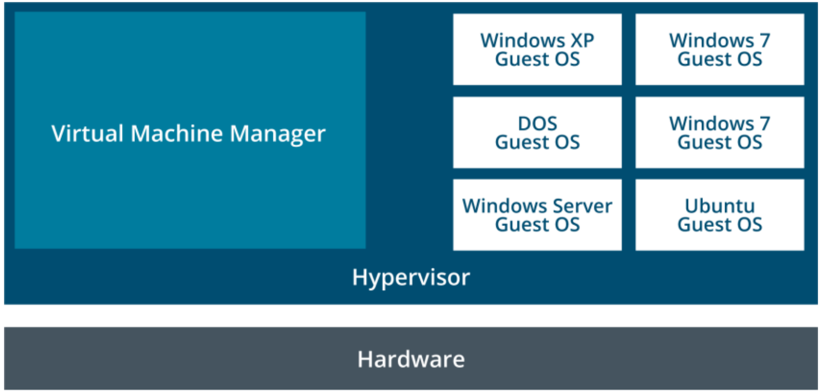
Type 1 (Bare metal): installed directly onto hardware

Hypervisors

TYPE 2 (HOST-BASED HYPERVISORS)



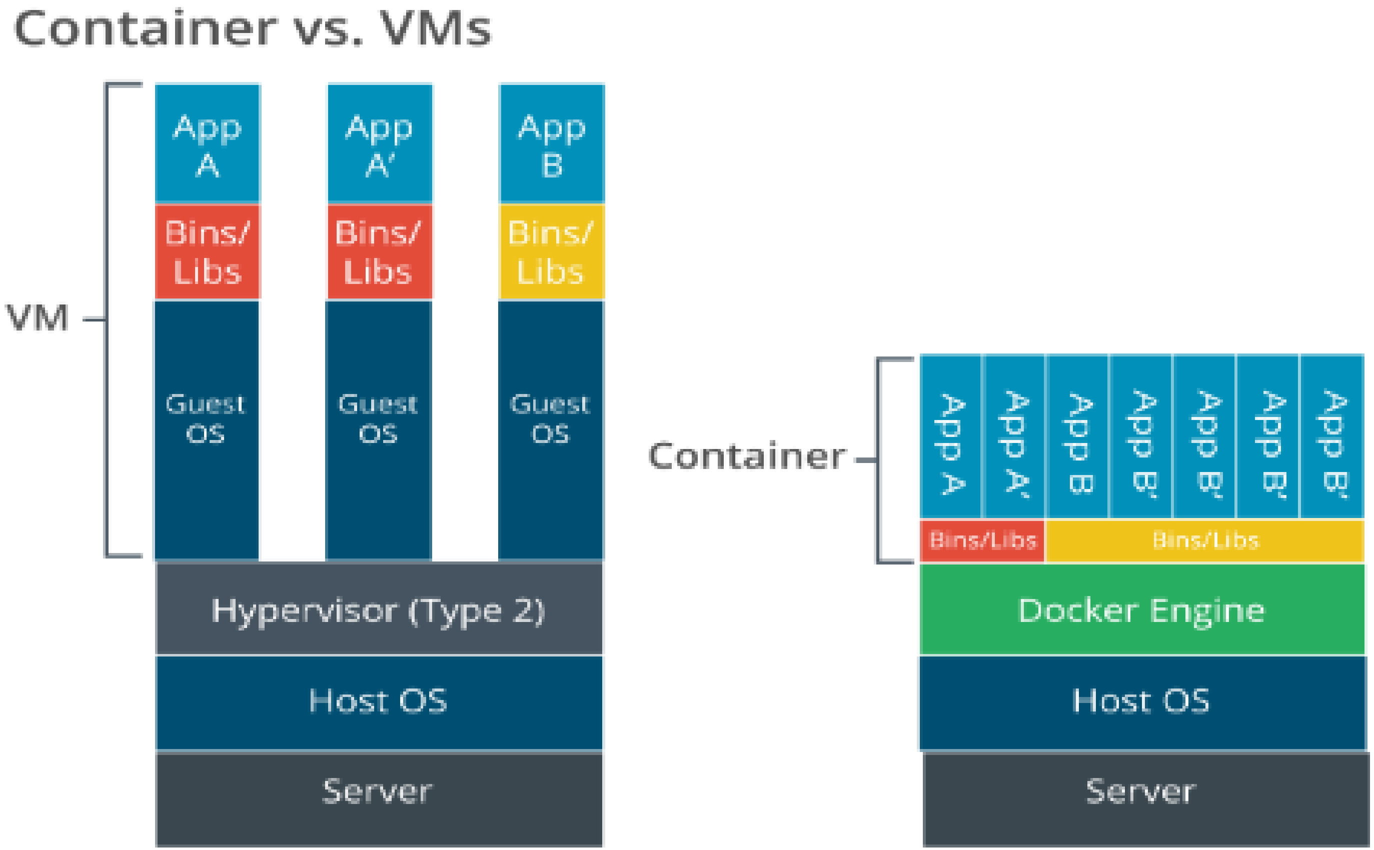
TYPE 1 (BARE METAL HYPERVISOR):



Uses for Virtualization

- Client-side virtualization
  - Sandboxing
  - Supporting legacy software and Oss
  - Cross-platform virtualization
  - Training
- Server-side virtualization
- Desktop virtualization
- Application virtualization
- Container virtualization

Comparison of virtual machines versus containers



Virtualization Resources Requirements

- CPU and virtualization extensions
- System memory for each OS guest
- Mass storage for each OS guest
- Networking for communication between VMs, host and other hosts

Virtualization Security Requirements

Guest OS security	Host security	Hypervisor security
• Use of template image for protection	• Host is a single point of failure needing extra protection	• Virtual machine escaping is a critical threat

Lab Activity

- Lab: Explore Virtualization
  (available in CertMaster Learn and CertMaster Perform)
- Lab: Create Virtual Hard Disks
  (available in CertMaster Learn and CertMaster Perform)

Lesson Review: Client-Side Virtualization

- Hypervisors
- Uses for Virtualization
- Virtualization Resource Requirements
- Virtualization Security Requirements

Lesson 8.2

Cloud Concepts

Learning Outcomes

What are the key characteristics of cloud computing that distinguish it from traditional on-premises IT infrastructure?

Which cloud deployment model would you recommend for a company that needs high availability and data security, and why?

How can Infrastructure as a Service (IaaS) and Software as a Service (SaaS) benefit a company in terms of resource provisioning and software development?

Learning Outcomes

What are some key advantages of using cloud file storage services like Microsoft OneDrive, Dropbox, Apple iCloud, and Google Drive for file synchronization and collaboration?

What are the three layers defined in the Software-Defined Networking (SDN) model by IETF, and what is the primary function of each layer?

Cloud Concepts

- Cloud Characteristics
- Common Cloud Deployment Models
- Common Cloud Service Models
- Cloud File Storage

Cloud Characteristics

Metered utilization model

Users pay only for services they use

Key benefits

- High availability
- Scalability
- Rapid elasticity

Shared resources pooling means datacenter hardware is for many rather than single customer

Common Cloud Deployment Models

Ownership and access arrangements

- Public (Multitenancy): offered by cloud service provider
- Private: cloud infrastructure exclusive to an organization
- Community: several organizations share private cloud
- Hybrid: combines public, private, and/or community elements

Common Cloud Service Models

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Infrastructure as a Service (IaaS)

- Organizations provision and manage IT resources

Software as a Service (SaaS)

- Applications delivered on Internet by subscription

Platform as a Service (PaaS)

- A development environment with infrastructure

Desktop as a Service (DaaS)

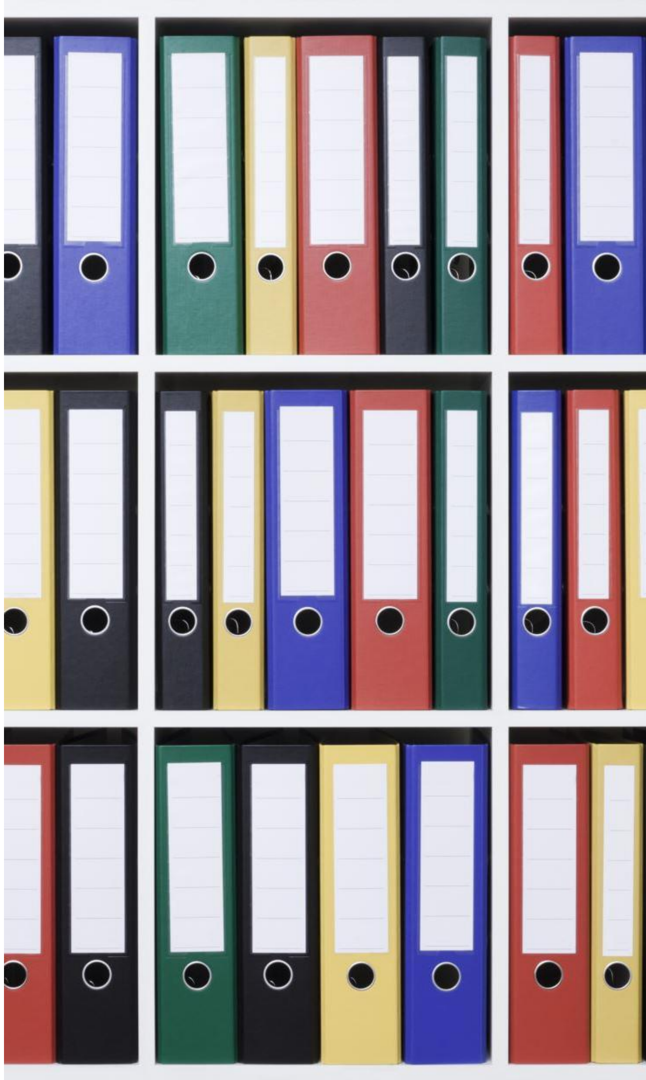
- Virtual desktops to end-users over the Internet

Cloud File Storage

A type of SaaS that allows users to store and sync files across many devices

Advantages

- Automated file synching
- Multiple users can collaborate on one document
- Fast access and reliability
- Available in different cost tiers



Lab Activity

- Use the Azure Interface
  (available in CertMaster Learn and CertMaster Perform)
- Manage IaaS Virtual Machines (VMs) in Azure That Run Windows Server
  (available in CertMaster Learn and CertMaster Perform)

Lesson Review: Cloud Concepts

- Cloud Characteristics
- Common Cloud Deployment Models
- Common Cloud Service Models
- Cloud File Storage